# Chayan Chuhan

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## Professional Summary

Data Science & AI/ML Engineer with hands-on experience in deep learning, predictive modeling, computer vision, and REST API development. Proficient in building end-to-end ML/DL pipelines covering EDA, feature engineering, model training, hyperparameter optimization, and deployment. Developed production-ready AI applications using Python, TensorFlow/Keras, Scikit-learn, and XGBoost, with deployments on Hugging Face Spaces.

## Technical Skills

- **Programming Languages:** Python, SQL
- **AI / Deep Learning:** TensorFlow, Keras, Scikit-learn, XGBoost, Bidirectional LSTM, RNN, NLP, Word2Vec, Optuna
- **ML Concepts:** Feature Engineering, EDA, Regression, Classification, $R^2$, MAE
- **Frameworks & APIs:** FastAPI, Streamlit, Pydantic, Uvicorn
- **Data & Analytics:** Pandas, NumPy, MySQL, Power BI, Cross-Validation
- **MLOps & Tools:** Git, GitHub, Hugging Face Spaces, Jupyter Notebook, VS Code

## Projects

***1. Product Reviews Sentiment Analysis using Bidirectional LSTM (NLP)***

- Built a multiclass sentiment analysis model using **205K+ Flipkart product reviews** to classify customer feedback as **Positive, Neutral, and Negative**.
- Preprocessed text data using tokenization, sequence padding, and pretrained **Word2Vec embeddings**, then trained a **Bidirectional LSTM** with dropout and Early Stopping to improve generalization.
- Achieved **~92% validation accuracy** and deployed the trained model with a reusable inference pipeline for predicting sentiment on new product reviews.

**Tech**: Python, TensorFlow, Keras, NLP, Bidirectional LSTM, Word2Vec, Scikit-learn, Pandas, NumPy

***2. Swiggy Delivery Time Prediction***

- Built an ML pipeline on **45K+ delivery records** — outlier handling, label encoding, and feature selection.
- Trained and tuned an XGBoost regressor; **achieved $R^2$ of 0.82** on held-out test data.
- Deployed on Hugging Face Spaces via Streamlit for live predictions.

**Tech**: Python, Scikit-learn, XGBoost, Streamlit, Hugging Face Spaces

***3. FastAPI Food Delivery Backend***

- Built REST APIs for menu, cart, and orders with delivery charge logic and Pydantic request validation.
- Handled CRUD, duplicate cart entries, availability guards, and checkout with grand total calculation.
- Added search, filtering (category, price, availability), sorting, and pagination across menu endpoints.

**Tech**: Python, FastAPI, Pydantic, Uvicorn

## Education

**B. Tech (IT)** – Aditya Silver Oak Institute of Technology, Ahmedabad | **CGPA:** 8.64/10 | 2021–2025

## Certifications

**Agentic AI Internship Certificate** – Innomatics Research Labs | **Python (Gold-Badge)** – HackerRank
**Machine Learning & Data Analytics** – Infosys Springboard